

Practice Lab:

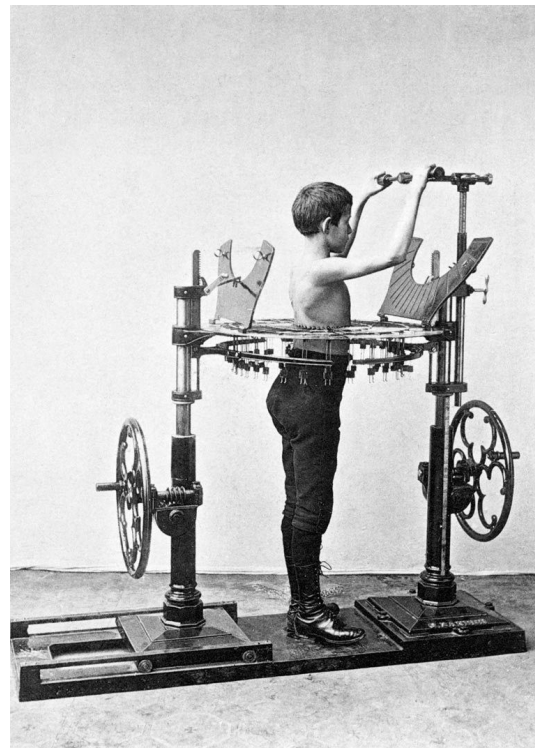
DSE Studio, Install, Configure, First Steps

- This Practice Lab is dependent on Discussion Unit 6210, where most of the objects we create in this lab were introduced.
- In this Practice Lab, we install DSE Studio from Tar ball, configure it, and run a basic install verification; create a table, add data, query same.

Challenge 1: Install, Configure, run DSE Studio

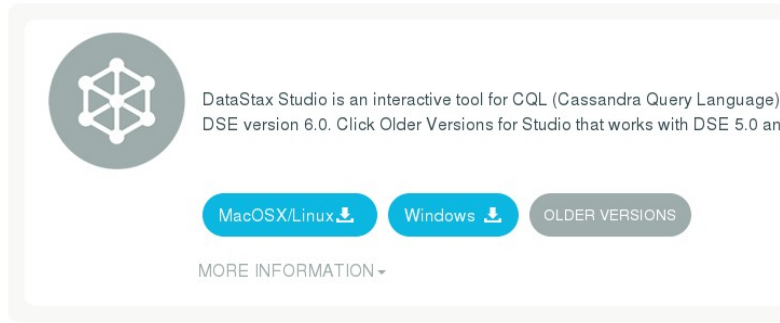
Prerequisites:

- Instructions are provided for Linux
- We will operate below recommended settings; 1 OS node only, minimum 8 GB RAM, 12 GB RAM preferred
- That you have an operating single node DSE Core cluster (anything more likely won't fit; DSE Core plus Studio on 8 GB RAM)
- All work done as 'root'



Challenge 1: Files, Directories

Studio



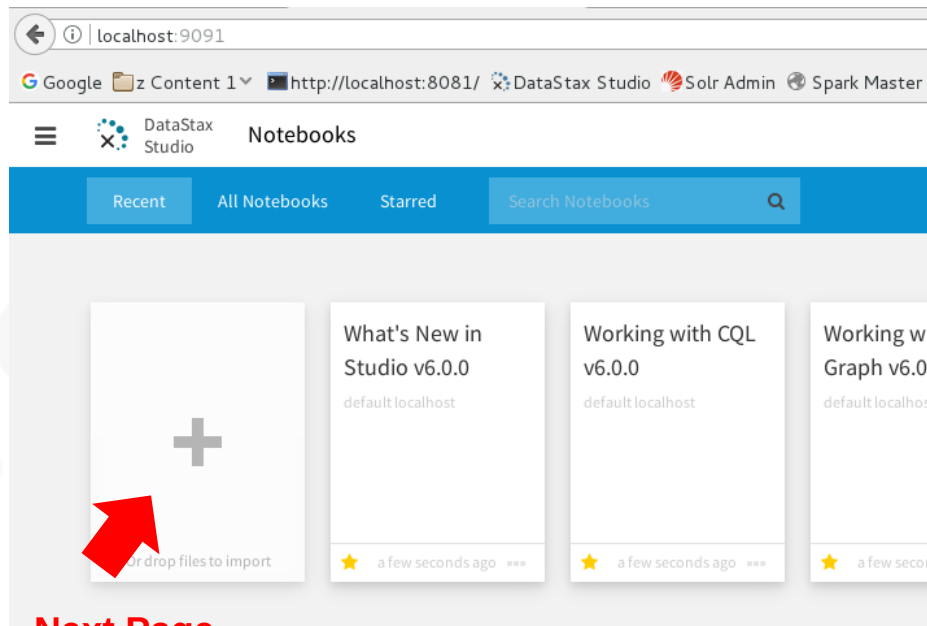
- Download Studio from Academy, <https://academy.datastax.com/downloads>
- Unpack the DSE Studio Tar ball in /opt/studio

Challenge 1: Configure

- `cd /opt/studio/conf`
- `ls -l`
- (Edit `configuration.yaml`)
 - Optional step, as no changes are required
 - We changed
 - `userData:`
 - `baseDirectory: /opt/studio_data`
 - Make sure directory above exists



Challenge 1: Start Studio (Web client server)



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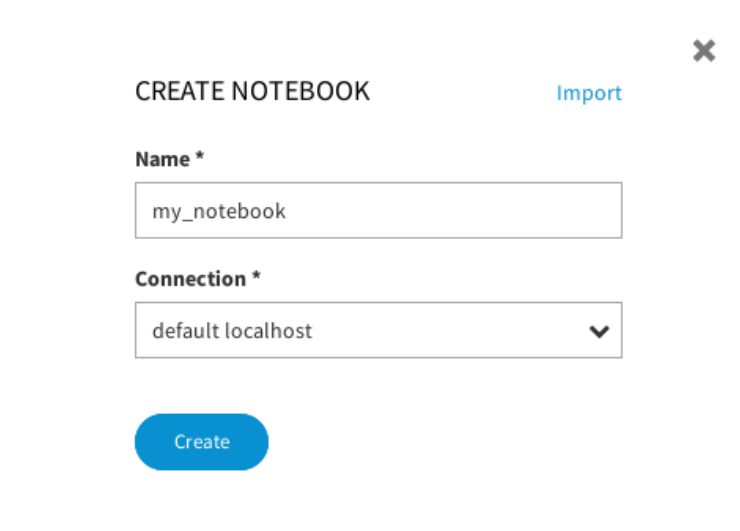
```
cd /opt/studio/bin  
./server.sh
```

In Web browser,
localhost:9091

Challenge 1: Install verify

In Studio Web UI:

- Click the large PLUS symbol
- This action produces a modal dialog box titled, CREATE NOTEBOOK
- Enter data as shown at right
- Click -> Create



CREATE NOTEBOOK [Import](#)

Name *

my_notebook

Connection *

default localhost

Create

Challenge 1: Make Keyspace, table, data, select

- Enter CQL to CREATE KEYSPACE
- Click the (Run button) (CL.ONE is a reference to consistency level one)
- Add a new Cell; Click the small grey plus symbol on the bottom-line/middle of any existing cell
- In the new Cell, chose a target keyspace
- Add and run CQL to CREATE TABLE, ..
- Run

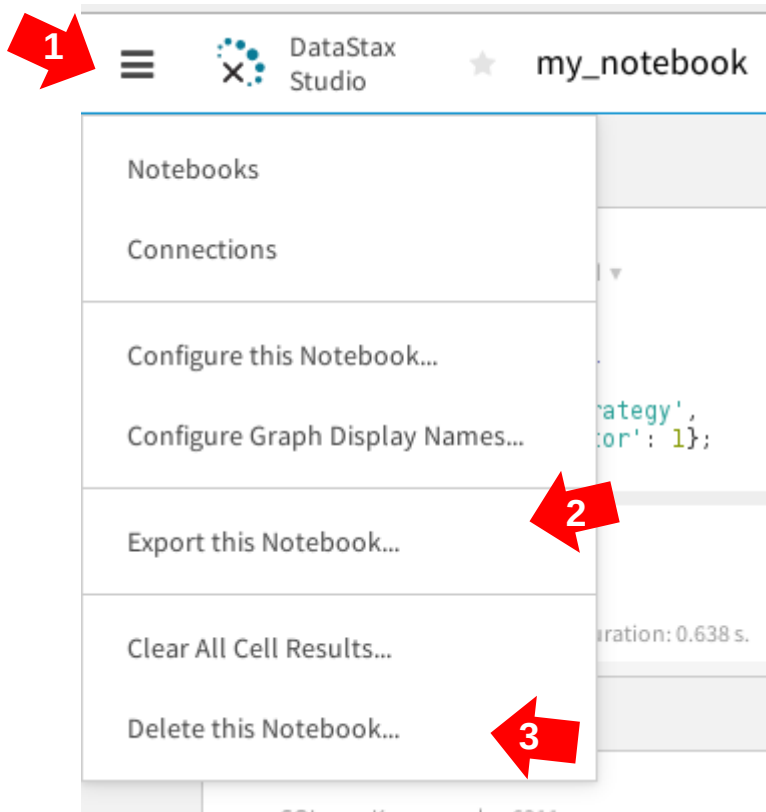
The screenshot shows the DataStax Studio interface. At the top, the CQL editor shows the command to create a keyspace: `CREATE KEYSPACE ks_6211 WITH REPLICATION = {'class': 'SimpleStrategy', 'replication_factor': 1};`. Below this, a message indicates "Success - No Data Returned" and "Success. 0 elements returned. Duration: 0.638 s.". A red arrow points from the CQL editor to the "CLONE" button. Below the "CLONE" button, another red arrow points to the CQL editor, which now shows the command to create a table: `CREATE TABLE t1 (col1 INT PRIMARY KEY, col2 TEXT); INSERT INTO t1 (col1, col2) VALUES (1, 'rrr'); INSERT INTO t1 (col1, col2) VALUES (2, 'sss'); SELECT * FROM t1;`. Below the CQL editor, there is a toolbar with icons for various views (code, table, pie chart, bar chart, line chart, area chart, scatter plot). The "table" icon is selected. Below the toolbar, a table is displayed with the following data:

index ↑
0
1

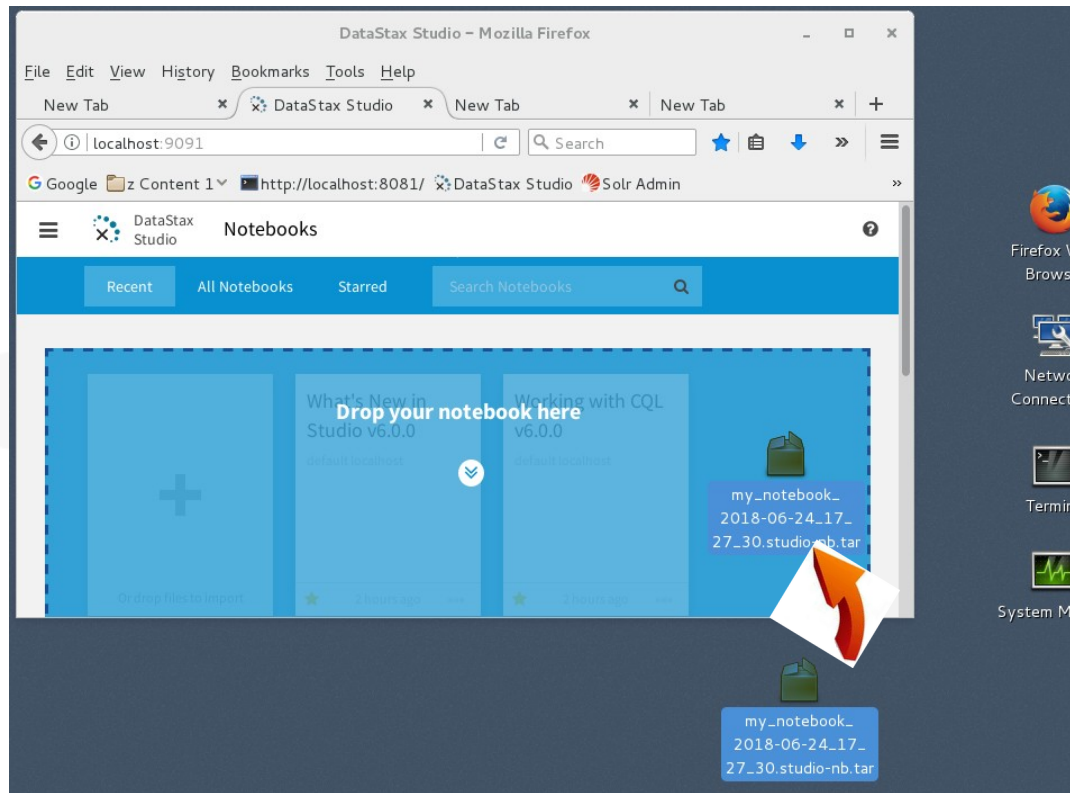
Challenge 2: Export/drop Notebook

In Studio Web UI:

- Click the left most icon in the Toolbar (balloon help titles this icon, Main Menu).
- Select -> Export
- Follow prompts to save (file, a Tar ball) to the Desktop.
- Then, follow prompts to "Delete this Notebook"



Challenge 2: Import a Notebook



In Studio Web UI:

- Drag and Drop the previously exported Notebook from the Desktop to the Studio (canvas)
- You will need to reselect Keyspaces on imported Notebooks

Go Father: (Optional)



Vasco Núñez de Balboa was a Spanish explorer (or conquistador) who saw and claimed the Pacific Ocean for Spain. He was born in 1475 in a poor region of Spain known as Extremadura, which was also the home of conquistadors Hernán Cortés, Francisco Pizarro, Hernando de Soto, and Francisco de Orellana.

Source: <http://mrnussbaum.com/explorers/famous-explorers/>

Challenge 3 (Optional): Run Spark/SQL

From the command line:

- Shutdown down your single node DSE Core
- Edit the `dse.yaml`. Change `always_on_sql_options:`
`enabled: true`

(With proper indentation)

- Restart DSE. Add a `-k` option on the command line
- Restart the DSE Studio daemon (see Notes)



Challenge 3 (Optional): Spurious error



Spark SQL ▾ Database: default ▾

```
|select * from ks_6211.t1;
```

System Error

Query execution exceeded configured execution timeout of 2147483647 seconds

Challenge 3 (Optional): Success

Spark SQL ▾ Database: default ▾

```
select * from ks_6211.t1;
```



index ↑
0
1

Displaying 1 - 2 of 2 results for the last statement

Challenge 3 (Optional): Definite Error (Resource)

Spark SQL ▾ Database: default ▾  Cell cannot be executed because DSE Spark Analytics workload is not enable
AlwaysON SQL service is not enabled and running in connection default loc

```
select * from ks_6211.t1;
```

System Error

The AlwaysOn SQL service is not available.

Practice Lab:

Lessons Learned



